Mars terrain type segmentation with novel edge detection

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***Abstract*—Utilizing deep learning techniques for Mars exploration can help ensure safer landing areas as well as avoid potentially dangerous areas for rover exploration [1]. In this research, we use Fully Convolution Networks [2] (FCN) to perform semantic segmentation. In addition, inspired by the DeepLabV3+ model [3] which uses an encoder-decoder structure to obtain a sharp object boundary, we add an additional edge channel to obtain the object boundary. Our result shows that adding an edge channel in the training phase could increase the recall rate of bedrock terrain types.**

***Index Terms*—semantic segmentation, mars terrain, FCN**

I. INTRODUCTION

Identifying terrain type is critical for rovers exploring planetary surfaces to prevent undue damage and success long after mission completion. In the case of the Mars rover Curiosity, which experienced wheel slip and sinkage due to the operation team commanding it to drive over rippled sand. Curiosity's wheels have also sustained significant damage while traversing over hard, undesirable terrain, such as bedrock.

For our research, we have chosen the AI4MARS version 0.3 as our dataset, as we believe computer vision can effectively address navigation issues. Further, the concept of identifying terrain type has far-reaching implications for future missions on various other celestial bodies such as Venus, which has a thick atmosphere that makes it challenging to capture satellite images of its terrain. This project not only satisfies our curiosity about space exploration but also helps us solve a practical problem in computer vision.

Our goal is to build upon existing research to improve the accuracy of terrain feature detection.

II. DATASET

The AL4MARS image dataset was created in collaboration with various space agencies, including NASA, ESA, and other international partners. The primary goal of the dataset is to foster research and development in the field of Martian surface analysis and to provide an extensive resource for researchers, educators, and students working on advanced algorithms for Martian exploration. The AI4MARS version 0.3 contains:

- MSL (Curiosity): 18,128 Images
- MER (Spirit & Opportunity): 16,301 Images

All images are grayscale, with three categories of labels, separated by the number of arguments for the pixels:

- 0: Soil
- 1: Bedrock
- 2: Sand
- 3: Big Rock
- 255: NULL

III. LITERATURE REVIEW OF EXISTING WORKS

Currently, there are two ways to obtain visual information from Mars: orbital and ground-based images. The 2016 Soil Property and Object Classifier (SPOC) paper [1] proposed using a training set that consisted of raw images (orbital or ground) and corresponding terrain labels provided by domain experts. However, due to the small training dataset and outdated CNN architecture, the results were not satisfactory. Despite this, the SPOC project aimed to incorporate deep learning techniques to enhance rover path decisions by including terrain type, instead of relying solely on classical perception systems based on geometric information.

To address the need for a larger training dataset, the AI4MARS project [4] leverages crowd-sourcing to collect labels. With a vast dataset and advanced CNN architecture, the module can achieve significantly better results than previous approaches. By utilizing these improved techniques, the AI4MARS project aims to enhance terrain identification and increase the accuracy of rover navigation on Mars. [5]

		10,000+ Volunteer Labels			
		Soil	Bedrock	Sand	Big Rock
Actual	Soil	99.10	0.32	0.57	0.01
	Bedrock	3.64	94.90	0.37	1.09
	Sand	0.88	5.62	93.45	0.05
	Big Rock	6.76	0	0	92.24
	Data			PA	mIoU
	1000 Labelled Images			0.5561	0.3528
	10,000+ Labelled Images			0.9667	0.9356

TABLE I
DEEPLABV3+ WITH A RESNET-101 BACKEND PRETRAINED ON IMAGENET APPROACH MIOU 0.93.

However, crowd-sourcing data is a luxury that isn't afforded when conducting a mission in real time. This limitation can potentially impact the model's accuracy over extended periods, especially in unfamiliar locations on Mars. As in table 3, with MER + MSL pretraining models have the highest mIOU, but it misclass' bedrock class to soil class. As such, a need to have a model that can, on its own, discern various feature types. As illustrated in table 2 below, no single model can accurately detect all features present on the planet.

	No Pretraining				
		Soil	Bedrock	Sand	Big Rock
Actual	Soil	59.57	14.19	25.47	0.03
	Bedrock	0.01	95.78	3.79	0.01
	Sand	0.16	4.67	93.45	0.01
	Big Rock	0.00	64.07	0	35.51

	ImageNet Pretraining				
		Soil	Bedrock	Sand	Big Rock
Actual	Soil	59.57	14.94	25.47	0.03
	Bedrock	0.01	95.78	3.79	0.01
	Sand	0.16	4.67	95.16	0.01
	Big Rock	0.00	64.07	0.42	35.51

	MER + MSL Pretraining				
		Soil	Bedrock	Sand	Big Rock
Actual	Soil	83.75	0.07	15.39	0.15
	Bedrock	28.13	53.56	11.71	6.6
	Sand	3.39	0.16	95.72	0.19
	Big Rock	6.98	27.08	1.91	64.03

TABLE II
CONFUSION MATRICES FOR M2020 TEST SETS WITH DIFFERENT PRETRAINING

Pretraining	PA	mIoU	mIoU w/o Big Rock
None	0.802	0.528	0.627
ImageNet	0.743	0.497	0.566
MER + MSL	0.842	0.544	0.673

TABLE III
M2020 QUANTITATIVE RESULTS.

IV. EXPERIMENT METHOD

We evaluated the semantic segmentation performance of Fully Convolutional Network (FCN) by comparing different sets of hyperparameters. Our goal was to accurately distinguish between different types of terrain. To enhance the detection of object boundaries, we incorporated an additional channel, the edge channel, into our input image.

To train our neural network, we only used the MSL dataset. To focus on terrain segmentation, we excluded data from areas that is far from 30-meter away, as well as the rover itself, from our training and evaluation.

V. EVALUATION METHODS

We use the same definition of pixel accuracy, recall, precision as this paper [5], but we include the background as fifth class. In cases where either the true positive count, false positive count, or false negative count is zero for a particular class during evaluation, the recall (or precision) metric may become undefined. To handle this, the np.nanmean function from NumPy can be used to calculate the average recall (or precision) across multiple classes while ignoring any NaN (not a number) values that may arise from 0/0 cases.

VI. RESULTS

Section 1: Comparing FCN performance with/without ImageNet pretrained weights and with additional edge channel

In Section 1, we compared two models, A and B, and found that using ImageNet pretrained weights did not affect the performance of our FCN model. This was good news for our future experiments because when adding edge channels, we couldn't apply any pretrained weights due to the modified CNN architecture. If using ImageNet pretrained weights had affected the performance, we would have had to pretrain the model ourselves.

Next, we compared models A and C and observed that the recall for bedrock type increased from 0.61 to 0.64, while the recall for soil type dropped.

Section 2: modify the hyperparameters to get better performance

In section 2, We made adjustments to the hyperparameters to improve performance, including increasing the batch size and lowering the learning rate. In this section, we fixed the fuzzy boundary caused by linear resizing.

When comparing models A and E, we observed an increase in the recall of soil and bedrock, as well as a significant increase in the precision of bedrock, sand, and bedrock. Similarly, when comparing models C and F (which included an additional edge channel), we also observed an increase in the recall of both soil and bedrock, and a notable increase in the precision of bedrock, sand, and bedrock.

Model Name	Batch	Learning Rate	Epoch	Pretrained	Edge	Recall, Precision, Pixel Accuracy: [Soil, Bedrock, Sand, Big Rock, Background]
A	16	0.001	80	True	False	R: [0.55, 0.61, 0.44, 0.07, 0.85] P: [0.81, 0.41, 0.30, 0.01, 0.83] PA: 0.84
B	16	0.001	80	False	False	R: [0.56, 0.60, 0.44, 0.07, 0.86] P: [0.79, 0.41, 0.33, 0.01, 0.83] PA: 0.84
C	16	0.001	80	False	True	R: [0.50, 0.64, 0.43, 0.07, 0.85] P: [0.83, 0.40, 0.32, 0.01, 0.81] PA: 0.83
D	48	0.0001	100	True	False	R: [0.59, 0.65, 0.44, 0.12, 0.85] P: [0.77, 0.67, 0.69, 0.15, 0.84] PA: 0.86
E	48	0.0001	80	True	False	R: [0.59, 0.65, 0.44, 0.08, 0.86] P: [0.79, 0.67, 0.76, 0.14, 0.84] PA: 0.86
F	48	0.0001	80	False	True	R: [0.57, 0.62, 0.42, 0.07, 0.85] P: [0.64, 0.61, 0.70, 0.15, 0.82] PA: 0.84

TABLE IV

VII. CONCLUSION

In this paper, we introduced the concept of using an edge channel to depict object boundaries and found that adding this channel helped improve the recall and precision of our model for Mars terrain type segmentation. Although our model is not yet comparable to state-of-the-art models, our results suggest that providing spatial information and object boundary is an effective approach for terrain type segmentation.

We also noted that the distribution of each terrain type in our training dataset was not balanced. To address this issue, we plan to use data augmentation techniques to increase the diversity of our dataset and balance the number of samples for each terrain type for our future work.

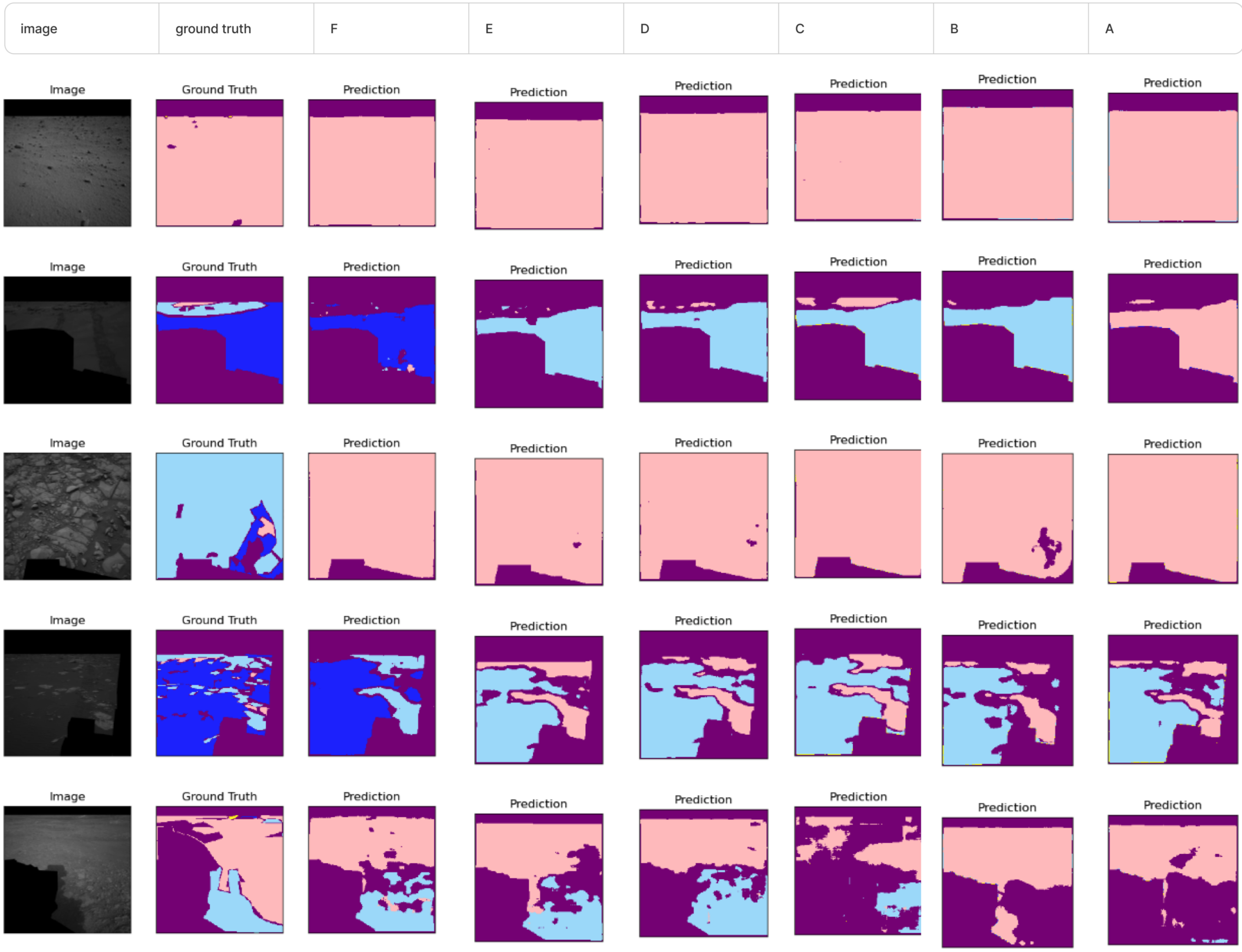

Fig. 1. Outputs of different models versus ground truth.

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